The Design and Implementation of a High Throughput Experimentation Platform to Accelerate Drug Discovery Supplemental Information

Amanda W. Dombrowski*, Ana L. Aguirre and Ying Wang

AbbVie, Inc., North Chicago, Illinois 60064, United States
amanda.dombrowski@abbvie.com

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Table S1. Success rate of each transformation

Transformation	#	Success Rate (%)
Amide coupling	9	76
C-N coupling	313	67
C-O coupling	77	55
Reductive cross electrophile coupling	3	30
Heck coupling	5	30
Late-stage fluorination	25	50
Late-stage methylation	19	45
Late-stage oxidation (chemical, biocatalytic)	39	78
Late-stage trifluoromethylation	1	0
Photoredox Copper sp^3 C-N coupling	4	100
Photoredox Nickel BF_3K coupling	31	77
Photoredox Nickel cross electrophile coupling	5	30
Photoredox Nickel C-N coupling	2	100
Photoredox Nickel decarboxylative coupling	39	55
Photoredox Nickel HAT C-H activation	1	100
Ring closing metathesis	4	44
Aromatic nucleophilic substitution	2	33
Sonogashira coupling	13	64
Sulfonamide formation	1	100
Suzuki coupling	208	78
Total	801	-

Table S2. Top Suzuki coupling reaction conditions by halide substructure

6-membered heterocyclic halides				
	Catalyst	**Base**	**Solvent**	**In top 10 list?**
1	CataCXium A Pd G3	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
2	Pd(DPEPhos)Cl_2	Cs_2CO_3	dioxane/H_2O (4:1)	
3	$PdCl_2(PPh_3)_2$	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
4	$PdCl_2$(dtbpf)	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
5	PEPPSI-IPr	KO*t*Bu	dioxane/H_2O (4:1)	
6-membered aryl halides				
	Catalyst	**Base**	**Solvent**	**In top 10 list?**
1	Xphos Pd G3	K_3PO_4	dioxane/H_2O (4:1)	Yes
2	Cataxium A Pd G3	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
3	Pd(Amphos)Cl_2	K_2CO_3	dioxane/H_2O (4:1)	Yes
4	$PdCl_2$(dtbpf)	K_2CO_3	dioxane/H_2O (4:1)	Yes
5	$PdCl_2$(dtbpf)	K_3PO_4	dioxane/H_2O (4:1)	Yes
6	Xphos Pd G3	Cs_2CO_3	dioxane/H_2O (4:1)	
5-membered heterocyclic halides				
	Catalyst	**Base**	**Solvent**	**In top 10 list?**
1	$PdCl_2$(dtbpf)	K_3PO_4	dioxane/H_2O (4:1)	Yes
2	Pd(Amphos)Cl_2	K_2CO_3	dioxane/H_2O (4:1)	Yes
3	$PdCl_2$(dtbpf)	K_2CO_3	dioxane/H_2O (4:1)	Yes
4	$P(tBu)_3$ Pd G3	Cs_2CO_3	dioxane/H_2O (4:1)	
5	Pd(dppe)Cl_2	K_2CO_3	dioxane/H_2O (4:1)	

Table S3. Top Suzuki coupling reaction conditions by boronate substructure

6-membered heterocyclic boronates				
	Catalyst	**Base**	**Solvent**	**In top 10 list?**
1	Pd(Amphos)Cl_2	K_2CO_3	dioxane/H_2O (4:1)	Yes
2	$PdCl_2$(dtbpf)	K_3PO_4	dioxane/H_2O (4:1)	Yes
3	$P(tBu)_3$ Pd G3	K_3PO_4	dioxane/H_2O (4:1)	Yes
4	Xphos Pd G3	K_3PO_4	dioxane/H_2O (4:1)	Yes
5	$PdCl_2$(dtbpf)	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
6-membered aryl boronates				
	Catalyst	**Base**	**Solvent**	**In top 10 list?**
1	$PdCl_2$(dtbpf)	K_3PO_4	dioxane/H_2O (4:1)	Yes
2	CataCXium A Pd G3	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
3	$PdCl_2$(dtbpf)	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
4	Pd(dppf)Cl_2-CH_2Cl_2	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
5	Pd(DPEPhos)Cl_2	Cs_2CO_3	dioxane/H_2O (4:1)	

5-membered heterocyclic boronates				
	Catalyst	**Base**	**Solvent**	**In top 10 list?**
1	$PdCl_2(PPh_3)_2$	Cs_2CO_3	dioxane/H_2O (4:1)	Yes
2	$PdCl_2(PCy_3)_2$	K_3PO_4	dioxane/H_2O (4:1)	
3	$PdCl_2(PPh_3)_2$	K_2CO_3	dioxane/H_2O (4:1)	
4	Xphos Pd G3	K_3PO_4	dioxane/H_2O (4:1)	Yes

Table S4. Top C-N coupling reaction conditions for 5-membered heterocyclic halides

	Catalyst	**Ligand**	**Base**	**Solvent**	**In top 10 list?**
1	RuPhos Pd G3	RuPhos	LiHMDS	dioxane	Yes
2	$Pd_2(dba)_3$	Me$_4$*t*BuXPhos	LiHMDS	*t*-amyl alcohol	Yes
3	BrettPhos Pd G3		Cs_2CO_3	dioxane	Yes
4	*t*BuXPhos Pd G3		P_2-Et	DMSO	
5	TrixiePhos Pd G3		Cs_2CO_3	dioxane	